

Novel Photonic Lantern with Higher Order Radial Mode Suppression

Jin Wei⁽¹⁾, Justin Vella⁽¹⁾, Andrew Ross-Adams⁽¹⁾, Christopher Betters⁽¹⁾, Barnaby Norris⁽¹⁾, Daniel Dahl⁽¹⁾, Sergio Leon-Saval⁽¹⁾

⁽¹⁾ Sydney Astrophotonic Instrumentation Laboratory, jin.wei@sydney.edu.au

Abstract We present the design and fabrication of a novel 6-core photonic lantern to suppress the LP_{02} mode, by omitting its central core. The device was characterised by off-axis digital holography achieving an average 27.41 dB mode suppression (versus 32 dB simulated) with -0.19 dB throughput loss.

Introduction

Photonic lanterns (PLs) are passive, monolithic optical devices that provide an adiabatic transition between single-mode, few-mode and multimode propagation [1]. Over the past decade, PLs have attracted substantial interest in a wide range of applications, including adaptive optics [2], spatial-division multiplexing in telecommunications [3], and beam shaping for high-power laser systems [4]. Furthermore, PLs with modal-selectivity have been demonstrated before by mixing waveguide types with different wave-guiding properties in the design [5]. Here, we go one step further and explore extreme mode suppression by purposely omitting cores altogether in the standard hexagonal array in PLs. We design, simulate and fabricate a 6-core multicore-fibre (MCF) PL and benchmark it against a conventional 7-core MCF PL (Fig. 1). The 6-core design intentionally removes the central core, thereby reducing the radial symmetry at the multimode end and modifying the modal-coupling condition. This geometric design selectively attenuates higher-order radial modes, in particular LP_{02} , without employing core-size variation or refractive-index detuning strategies commonly used to lift modal degeneracy in mode-selective lanterns [2]. Suppression of LP_{02} mitigates crosstalk into other LP modes (e.g., LP_{11} and LP_{21}), thereby improving modal fidelity [6]. This is beneficial for the preparation and transmission of spatial-mode states, including orbital angular momentum (OAM) modes, which constitute an important mode basis in quantum photonics [7, 8] and are particularly promising for higher-dimensional quantum key distribution (QKD) [3]. Overall, our simulation and experimental results indicate that the missing-core hexagonal geometry offers a simple and effective route to intrinsic radial-mode filtering in PL architectures.

Theory and Method

We used Rsoft [9] to simulate two PL designs: one based on a standard hexagonal 7-core multicore fibre (MCF), and another utilising a 6-core MCF

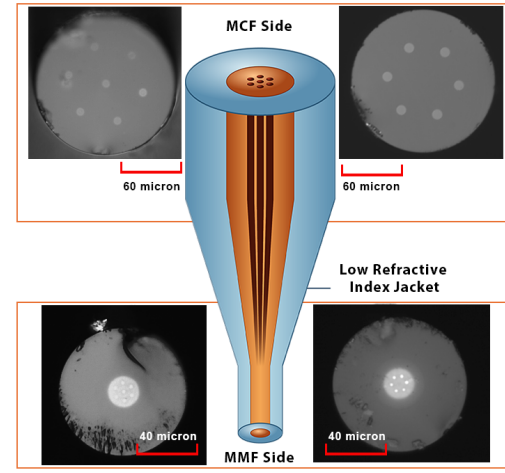


Fig. 1: Comparison between the core configurations of the two PLs: Left: 7-core MCF PL, showing the untapered 7-core MCF input alongside its MMF output with a diameter 24.37 μm . Right: 6-core MCF PL with the untapered 6-core MCF input and its MMF output with a diameter of 18.3 μm . The centre presents a schematic illustration of the 7-MCF PL.

with identical parameters, except for the omission of the central core. The FemSIM module was used to model the supported LP modes at the multimode (MM) end of each PL. The supported LP modes were injected into both simulated structures, enabling the analysis of the evolution of their respective effective refractive indices, N_{eff} , as depicted in the left panel of Fig. 2. The simulation results in Fig. 2 indicate that both PLs are capable of supporting the first six LP modes, spanning from LP_{01} to LP_{02} , as evidenced by their final N_{eff} values exceeding the cladding refractive index (represented by the dashed lines) at the single-mode (SM) ends of the PLs. Port monitors within the BeamPROP module were used to compute the mode-field distributions at the end of the transition, enabling precise characterisation for both lantern designs. The simulated electric field distributions served as the foundation for constructing the final transformation matrix for each PL, which is crucial for understanding the modal mapping between the MM and single-mode (SM) ends. The resulting transformation matrices are displayed in the

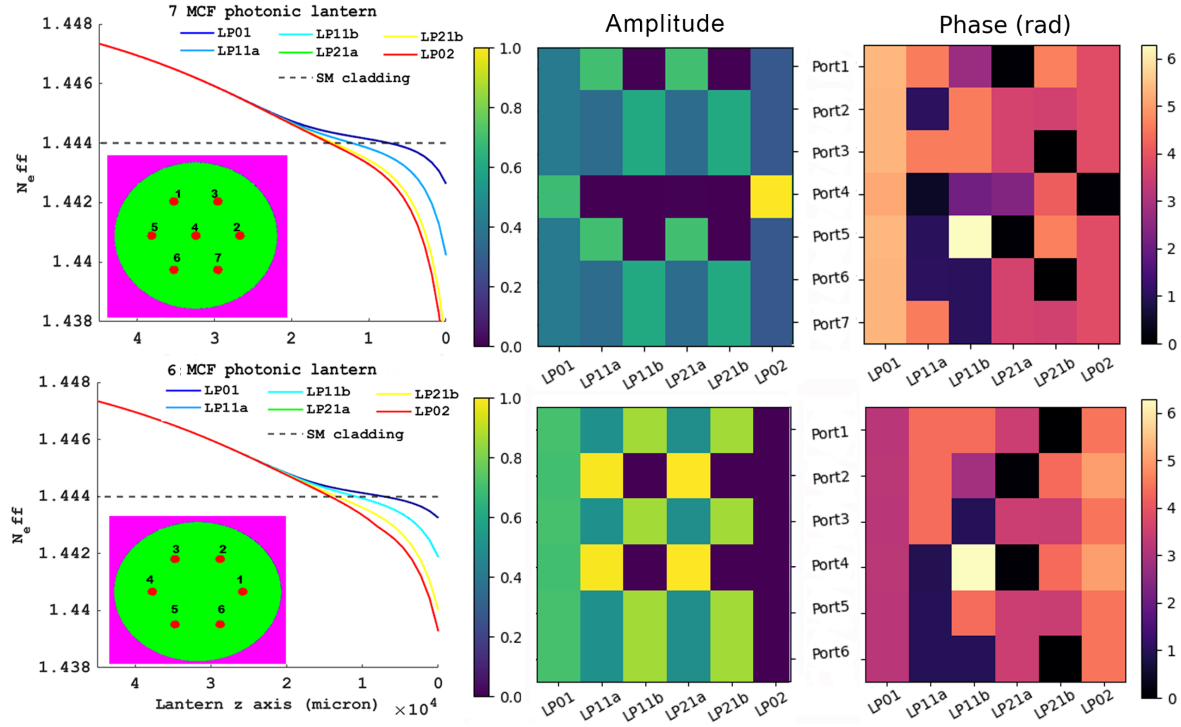


Fig. 2: Comparison between the 6-MCF PL and 7-MCF PL and their respective N_{eff} evolutions along the z axis: Left panels show the simulated evolution of the effective refractive index, N_{eff} , along the lantern transition (z axis) for supported LP modes in (top) a 7-core MCF PL and (bottom) a 6-core MCF PL. The numbers assigned to the PLs' cores correspond to the port order shown on the right side of the panel. Right side of the panel shows the transformation matrices, one for amplitude and one for phase, for the 7-MCF PL (top right) and the 6-MCF PL (bottom right).

right column of Fig. 2, providing a comprehensive overview of the mode conversion characteristics for both the standard hexagonal 7-core and the 6-core PL designs.

Concurrently, a custom 6-core MCF with an intentionally omitted central core was inserted into a low-refractive-index capillary and tapered down using a 3SAE Combiner Manufacturing System [10], a computer-controlled tapering machine. The resulting PL had a final numerical aperture (NA) of 0.12 and an MM-side diameter of $18.3 \mu\text{m}$. A second PL, fabricated from a standard telecommunications 7-core MCF and tapered under identical conditions (Fig. 1), produced a final MM-side diameter of $24.37 \mu\text{m}$ with the same NA (0.12). Using a narrow-band 1550-nm laser as the test source, both 6-MCF PLs exhibited a throughput of -0.19 dB, whereas the 7-MCF PL showed a throughput of -0.45 dB. The observed throughput difference of 0.25 dB was attributed to run-to-run fabrication variability. The diameters on the MM side of both PLs were fabricated above the LP_{02} cutoff, consistent with the N_{eff} curves shown in Fig. 3. To facilitate straightforward integration into the experimental test bench, both devices were mounted in a custom holder. An experimental test bench was implemented to characterise the PL using off-axis digital holography as explained in [11]. A fundamental-mode reference beam was interfered with the optical field emerging from the PL MM output, enabling simultaneous recovery of the am-

plitude and phase of the MM field [12, 13]. The resulting interferogram was used to reconstruct the complex electric field at the MM side, from which the modal decomposition was obtained and the purity of the targeted LP modes was evaluated [14, 15].

Results

The simulations show that the 6-MCF photonic PL exhibits pronounced suppression of the LP_{02} mode (Fig. 2). Mode-resolved extinction ratios (ERs) were extracted from the corresponding transformation matrices and are compared for the 6-MCF and 7-MCF designs in the left column of Fig. 3. Here, the ER is defined as the ratio of the power in the LP modes, from LP_{01} to LP_{21} , to the power in the undesired mode, LP_{02} . The 6-MCF PL achieves an average ER of 32 dB, indicating that, although the 6-MCF geometry could in principle support and guide the 6 LP modes at the multimode input, the LP_{02} mode is not preserved through the PL transition. By contrast, the conventional seven-core hexagonal configuration does not suppress LP_{02} .

Experimentally, the transformation matrices of the 6- and 7-MCF PLs were measured (Fig. 3) to evaluate the excitation and transfer behaviour of the LP_{02} mode in each configuration. As shown in the bottom row of Fig. 3, the 6-MCF PL exhibits consistently weak coupling of the LP_{02} mode across most input ports, indicating poor modal

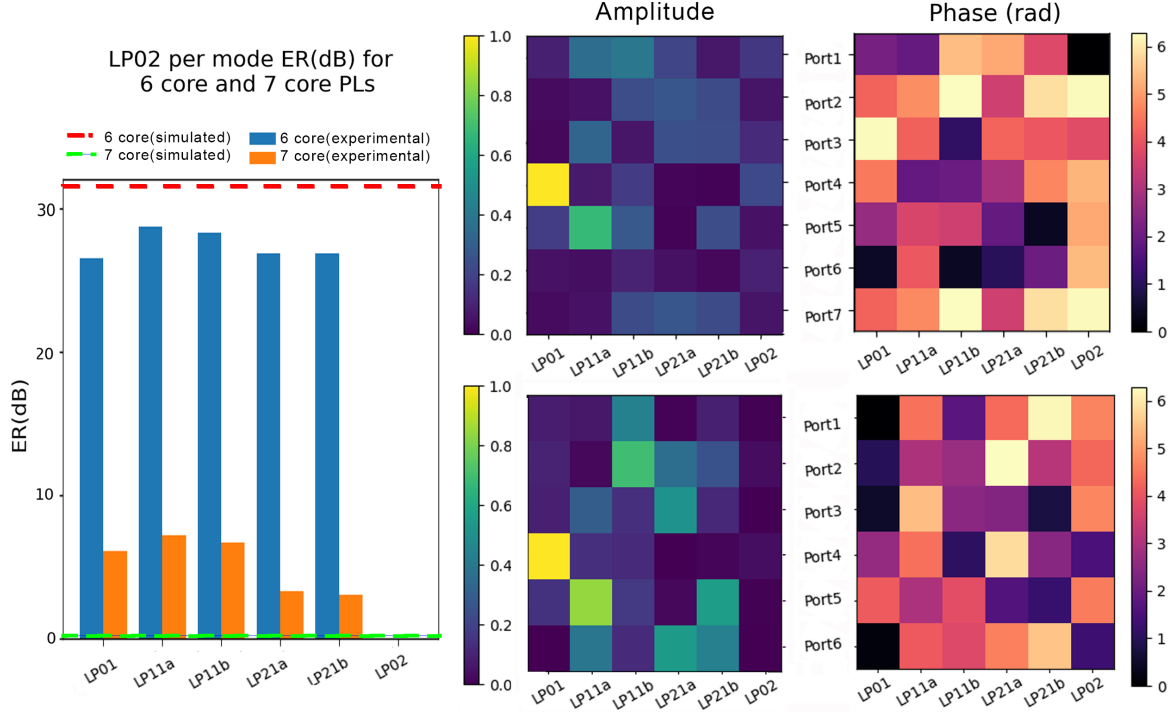


Fig. 3: Comparison in power suppression of LP_{02} against the first 6 LP modes that are supported by the MM side in both 6-MCF PL and 7-MCF PL: The experimental results show a clear divergence in LP_{02} suppression between the 6-MCF and 7-MCF photonic lanterns. On the left side of the figure, the extinction ratio (ER) extracted from the experimentally derived transformation matrix indicates strong suppression of LP_{02} in the 6-MCF device. The dashed red line represents the average ER from simulations for the 6-MCF PL, while the dashed green line corresponds to the simulated average for the 7-MCF PL. The 6-MCF PL (blue) maintains LP_{02} suppression of 32 dB across modes from LP_{01} to LP_{21b} . In contrast, the 7-MCF PL (orange), evaluated under identical conditions, exhibits no measurable suppression of LP_{02} . Also on the left side of the figure, the top row presents the amplitude and phase transformation matrices for the 7-MCF PL, while the bottom row shows the corresponding amplitude and phase matrices for the 6-MCF PL.

preservation through the lantern. In contrast, the 7-MCF PL (top row of Fig. 3) displays markedly stronger LP_{02} coupling. A quantitative comparison of the extinction ratio (ER) relative to the LP_{02} mode (Fig. 3), extracted from the transformation matrices, further highlights this distinction. The 6-MCF PL achieves an average per-mode suppression of 27.41 dB in power, whereas the 7-MCF PL provides only approximately 5.25 dB suppression value on average, which is a number coherent with the standard mode-mixing and evolution in a standard geometry PL. Furthermore, the limited suppression observed in the latter is more plausibly attributed to mode-dependent loss and fabrication non-idealities, rather than intentional modal discrimination. Overall, these experimental observations are consistent with theoretical predictions from RSoft simulations, confirming that removal of the central core in a hexagonal geometry is essential for effective suppression of higher-order radial modes.

Conclusion

Our results confirm that the 6-MCF PL geometry is a key determinant in suppressing higher-order radial modes. In on-bench measurements, the 6-MCF PL achieved an average LP_{02} suppression of 27.41 dB relative to the conventional 7-MCF

PL. This value is consistent with the 32 dB suppression predicted by simulations using the same geometrical parameters. Such intrinsic rejection of higher-order radial content is particularly relevant for waveguide platforms that demand strong attenuation of modes such as LP_{02} , including systems used for generating orbital angular momentum (OAM) states for quantum photonics applications, such as higher-dimensional QKD.

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References

- [1] S. G. Leon-Saval, T. A. Birks, J. Bland-Hawthorn, and M. Englund, "Multimode fiber devices with single-mode performance", *Optics Letters*, vol. 30, no. 19, pp. 2545–2547, Oct. 2005. DOI: 10.1364/OL.30.002545.
- [2] B. Norris, C. Betters, J. Wei, S. Yerolatsitis, R. Amezcua-Correa, and S. Leon-Saval, "Optimal broadband starlight injection into a single-mode fibre with integrated photonic wavefront sensing", *Opt. Express*, vol. 30, no. 19, pp. 34 908–34 917, Sep. 2022. DOI: 10.1364/OE.465639. [Online]. Available: <https://opg.optica.org/oe/abstract.cfm?URI=oe-30-19-34908>.
- [3] G. Milione, T. A. Nguyen, J. Leach, D. A. Nolan, and R. R. Alfano, "Higher-dimensional orbital-angular-momentum-based quantum key distribution with mutually unbiased bases", *Physical Review Letters*, vol. 115, no. 19, p. 190 501, 2015. DOI: 10.1103/PhysRevLett.115.190501.
- [4] J. Montoya, C. Hwang, D. Martz, C. Aleshire, T. Y. Fan, and D. J. Ripin, "Photonic lantern kw-class fiber amplifier", *Opt. Express*, vol. 25, no. 22, pp. 27 543–27 550, Oct. 2017. DOI: 10.1364/OE.25.027543. [Online]. Available: <https://opg.optica.org/oe/abstract.cfm?URI=oe-25-22-27543>.
- [5] A. M. Velázquez-Benítez et al., "Scaling photonic lanterns for space-division multiplexing", *Scientific Reports*, vol. 8, no. 1, p. 8897, 2018. DOI: 10.1038/s41598-018-27072-2.
- [6] D. J. Richardson, J. M. Fini, and L. E. Nelson, "Space-division multiplexing in optical fibres", *Nature Photonics*, vol. 7, no. 5, pp. 354–362, 2013. DOI: 10.1038/nphoton.2013.94.
- [7] D. N. Matsukevich and A. Kuzmich, "Quantum state transfer between matter and light", *Science*, vol. 306, no. 5696, pp. 663–666, 2004. DOI: 10.1126/science.1103346. eprint: <https://www.science.org/doi/pdf/10.1126/science.1103346>. [Online]. Available: <https://www.science.org/doi/abs/10.1126/science.1103346>.
- [8] Z. Zhou et al., "Orbital angular momentum photonic quantum interface", *Light: Science & Applications*, vol. 5, no. 4, e16019, 2016. DOI: 10.1038/lsa.2016.19.
- [9] RSoft Design Group, *Rsoft photonic device tools, version 2024.09*, Computer software, Synopsys, Inc. Mountain View, CA, USA, 2024.
- [10] 3SAE Technologies Inc. "3SAE combiner manufacturing system (cms)", Accessed: Apr. 20, 2026. [Online]. Available: <https://3sae.com/glass-processing/combiner-manufacturing-system/>.
- [11] A. K. Taras et al., "Illuminating the lantern: Coherent, spectro-polarimetric characterization of a multimode converter", *Opt. Express*, vol. 34, no. 1, pp. 1012–1025, Jan. 2026. DOI: 10.1364/OE.583186. [Online]. Available: <https://opg.optica.org/oe/abstract.cfm?URI=oe-34-1-1012>.
- [12] J. Carpenter, B. C. Thomsen, and T. D. Wilkinson, "Digital holography for modal content analysis of optical fibres", *Journal of Lightwave Technology*, vol. 30, no. 2, pp. 198–206, 2012. DOI: 10.1109/JLT.2011.2174989.
- [13] M. Takeda, H. Ina, and S. Kobayashi, "Fourier-transform method of fringe-pattern analysis for computer-based topography and interferometry", *Journal of the Optical Society of America*, vol. 72, no. 1, pp. 156–160, 1982. DOI: 10.1364/JOSA.72.000156.
- [14] J. Carpenter, B. J. Eggleton, and T. D. Wilkinson, "Complete modal decomposition of optical fields using digital holography", *Optics Letters*, vol. 39, no. 2, pp. 231–234, 2014. DOI: 10.1364/OL.39.000231.
- [15] J. Carpenter, *Digholo : High-speed library for off-axis digital holography and hermite-gaussian decomposition*, 2022. arXiv: 2204.02348 [eess.IV]. [Online]. Available: <https://arxiv.org/abs/2204.02348>.